A Low-Cost Conductive Grid System for Real-Time Multi-Metric Braille Reading Performance Analysis

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***Abstract*—Objective assessment of Braille reading proficiency remains essential yet underserved, with existing tools either prohibitively expensive or unsuitable for widespread classroom deployment. This paper presents a scalable, low-cost conductive grid–based sensing system that detects multi-touch interactions over a 7×7 Braille sensing grid in real time. An ESP32 microcontroller drives a multiplexed row–column scanning architecture, streaming capacitively coupled touch frames to a host computer over USB–serial. A multi-threaded Python pipeline computes seven complementary performance metrics: rolling Words per Minute (WPM) with exponential moving average smoothing, velocity profile inter-quartile range (IQR) as a motor-consistency indicator, inter-word regression rate and hesitation analysis, word-level Time-on-Task (ToT), grid coverage and skip-rate analysis, path efficiency with LOWESS trend fitting, and a composite difficulty score that unifies all signals into a per-word indicator. The system operates without any wearable device or optical apparatus, providing an unobtrusive, low-cost platform for diagnosing Braille reading proficiency in classroom and research settings.**

***Index Terms*—Braille reading, conductive grid, touch sensing, capacitive coupling, ESP32, finger tracking, WPM, time-on-task, regression analysis, path efficiency, motor consistency, accessibility, low-cost assistive technology**

I. Introduction

Braille literacy remains a cornerstone of independence and academic achievement for individuals who are blind or visually impaired. Research consistently demonstrates that Braille-literate individuals enjoy substantially higher employment rates [10]—the National Federation of the Blind reports that Braille-literate blind adults constitute approximately 90% of employed blind individuals [3]. The World Health Organization further reports that vision impairment carries an annual global productivity loss of approximately US$ 411 billion [11], underscoring the economic imperative of effective tactile literacy programmes. Yet despite this importance, tools for *objectively* measuring and improving Braille reading skills remain scarce, expensive, and inaccessible to most educators and learners.

Knowlton and Wetzel [12] reported a mean Braille reading speed of 136 words per minute (WPM) for experienced readers in controlled reading tasks, with a range of 65–185 WPM. The National Federation of the Blind separately reports that highly proficient two-handed readers can reach up to 200–400 WPM under optimal conditions [3], reflecting the upper tail of expert performance rather than typical experienced-reader speeds. However, beginners typically read at only 50–70 WPM. Understanding *why* a reader is slow—whether due to uneven finger movement, frequent backtracking, or inefficient reading patterns—requires instrumentation that captures more than aggregate reading speed. Existing systems for analysing Braille reading behaviour rely on vision-based tracking, wearable sensors, or dedicated electronic displays, all of which are intrusive, expensive, or impractical for widespread deployment.

There is therefore a pressing need for a low-cost, non-invasive system capable of capturing real-time tactile interaction data—including finger movement patterns, per-cell reading durations, total reading time, and average reading rate—without instrumenting the reader. This paper presents such a system. A passive conductive grid placed beneath a 7×7 Braille sensing grid detects finger contact events through capacitive coupling; a multi-threaded software pipeline then computes seven complementary metrics that together profile both reading fluency and motor consistency. The key contributions of this work are:

- A low-cost, non-invasive conductive grid system capable of localising touch interactions on a physical Braille page in real time.
- A real-time reading speed estimation framework (WPM) for monitoring Braille reading fluency.
- A movement-consistency analysis module (velocity IQR) for evaluating the stability of finger motion during reading.
- A regression and hesitation detection mechanism for identifying rereading behaviour and potentially difficult words.
- A time-on-task analysis framework that measures cumulative contact duration on individual word regions.
- A coverage analysis method for detecting skipped or insufficiently explored grid areas.
- A path-efficiency metric for assessing reading progression, and a composite difficulty score that aggregates all per-word metric signals into a single actionable indicator.

The remainder of this paper is organised as follows. Section II surveys existing Braille reading analysis technologies and identifies their limitations. Section III describes the hardware design, firmware architecture, and software pipeline. Section IV presents experimental results for each metric. Section V discusses observed reading behaviour patterns. Section VI outlines limitations and directions for future work, followed by conclusions in Section VII.

II. Related Work

Prior approaches to Braille reading analysis can be broadly categorised into four classes: marker-based tracking systems, wearable sensor systems, software-based tablet solutions, and vision-based and electronic approaches. This section reviews representative systems from each category and identifies the gaps that motivate the present work.

A. Marker-Based Tracking Systems

Early Braille-reading studies relied on external tracking hardware to capture finger motion. Hughes [5] used a digitising tablet to analyse Braille reading kinematics, revealing frequent velocity fluctuations and smoother motion profiles among skilled readers. Similarly, Aranyanak and Reilly [6] developed a Wiimote-based infrared tracking system that monitored LEDs attached to readers' fingers while supporting both embossed Braille and refreshable displays. Although these systems provide valuable movement data, they require markers to be attached to the reader, careful calibration, and specialised hardware that limits practical classroom deployment.

B. Wearable Sensor Systems

Several wearable approaches have been proposed for Braille reading assistance and analysis. Wii Braille [9] adapts Nintendo Wii components for finger-motion tracking, while the Finger-Eye system [2] combines a miniature camera with electro-tactile feedback to guide users during reading. While these approaches can achieve high precision, they require users to wear and calibrate additional hardware, potentially altering the natural reading experience.

C. Software-Based Tablet Solutions

Rassmus-Gröhn [4] proposed a tablet-based finger-tracking system in which transparent Braille paper is placed over a capacitive touchscreen. The system records finger movement patterns, reading durations, and reading speed while preserving a largely non-invasive interaction. However, it is limited to specific tablet hardware, supports only two-finger tracking, and constrains the reading area to the dimensions of the touchscreen.

D. Vision-Based and Electronic Approaches

Refreshable Braille displays with integrated sensing capabilities are commercially available but are often expensive and restricted to electronic content [3]. Vision-based approaches use cameras and computer-vision algorithms to track finger movement [1], but are sensitive to lighting conditions, occlusion, reader posture, and privacy concerns, making reliable deployment challenging.

E. Summary of Gaps

A common limitation across prior work is the extraction of only a limited set of performance metrics, typically reading speed or error rate. No existing low-cost system simultaneously tracks velocity consistency, word-level dwell time, regression patterns, skip rate, and path efficiency within a unified real-time pipeline. The proposed system addresses this gap.

TABLE I
Comparison of Braille Reading Analysis Systems

System	Low Cost	Non-Invasive	Real-Time	Vision Free
Marker-based [5], [6]	✓	×	✓	✓
Wearable [8], [9]	✓	×	✓	✓
Tablet touch [4]	✓	✓	✓	✓
Vision-based [1]	✓	✓	✓	×
Braille display [3]	×	✓	✓	✓
Proposed	✓	✓	✓	✓

III. Proposed System

A. System Overview

The proposed system consists of three components: (i) a passive conductive grid embedded beneath a Braille reading surface, with software-defined mapping between Braille content and grid coordinates; (ii) an ESP32 microcontroller board that drives multiplexed row–column scanning and streams capacitively coupled touch frames to a host PC over USB–serial; and (iii) a multi-threaded Python software pipeline that computes seven reading performance metrics in real time and renders a six-panel live visualisation suite. No modifications to the Braille page or the reader's hands are required.

Most Braille readers use only two fingers—typically the left and right index fingers—as the primary reading digits [4]. The conductive grid detects aggregate touch contact without requiring per-finger identification, which is sufficient for computing all seven proposed metrics. The system can therefore capture characteristic reading strategies such as the butterfly and side-by-side patterns (Section V) through spatial and temporal analysis of touch sequences.

B. Hardware Design

1) Conductive Grid Architecture: The sensing layer is a regular matrix of conductive copper-strip traces arranged in a 7×7 row–column configuration, providing $N = 49$ spatially distinct sensing locations. These can be mapped in software to Braille characters, words, or larger reading regions depending on the target application. In the current prototype, each sensing location corresponds to a single word position. The physical layout is shown in Fig. 1.

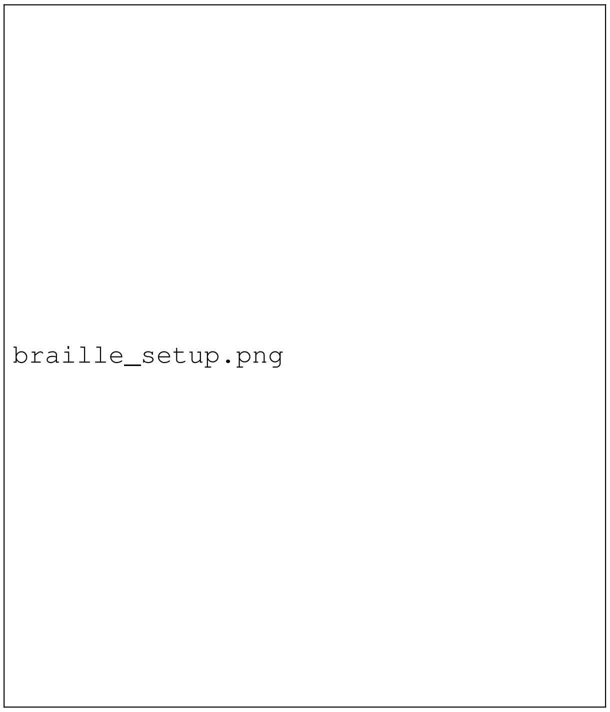

Fig. 1. Physical layout of the 7×7 conductive grid with ESP32 controller board.

2) Excitation Signal and Scanning: A hardware timer generates a $\sim$100 kHz square-wave excitation signal on a dedicated GPIO output pin. This AC carrier is routed through each selected transmit (TX) row in turn; the capacitively coupled signal is detected on the receive (RX) column lines when a conductive finger bridges a row–column intersection through the Braille page.

A 3-bit multiplexer selects one TX row at a time. For each selected row, all seven RX column lines are sampled sequentially. Each ADC reading averages 30 successive 12-bit samples; a per-channel settling delay is inserted after each multiplexer switch to allow the RC network to stabilise. This procedure yields a complete 7×7 activation frame per scan cycle, transmitted to the host over USB–serial. Key hardware parameters are summarised in Table II.

TABLE II
ESP32 HARDWARE CONFIGURATION

Parameter	**Value**
Microcontroller	ESP32-WROOM-32
Clock speed	240 MHz
ADC resolution	12-bit SAR
ADC input range	0–3.3 V
Excitation frequency	$\sim$100 kHz square wave
Samples per ADC read	30 (averaged)
MUX settling delay	25 μs per channel
Serial baud rate	115,200 baud
Grid dimensions	7×7 (49 cells)

3) Controller Board: The firmware orchestrates the complete scan–filter–transmit cycle, with threshold filtering applied in the host-side software pipeline to reject electromagnetic interference. Frame timestamps are derived from the microcontroller's microsecond-resolution hardware timer to support latency measurement.

C. Firmware Flowchart

Fig. 2 illustrates the ESP32 firmware control flow. After initialisation, the main loop iterates over all seven TX rows, selects each via the multiplexer, waits for RC settling, and reads all seven RX columns with a 30-sample averaged ADC filter. The resulting 7×7 activation matrix is transmitted to the host.

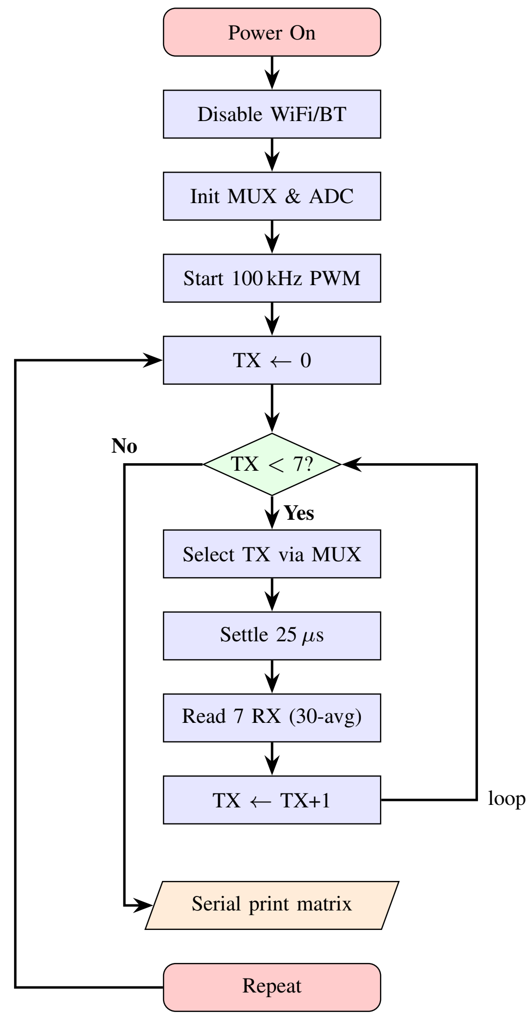

Fig. 2. ESP32 firmware control flow: multiplexed row scanning with filtered ADC readout and serial transmission.

D. Software Pipeline

The host-side pipeline runs two concurrent threads: a *metrics thread* that ingests raw serial frames and updates all tracker objects, and a *UI thread* that reads metric snapshots and renders the visualisation suite. All tracker classes expose a thread-safe `snapshot()` method protected by `threading.Lock`. The pipeline is illustrated in Fig. 3.

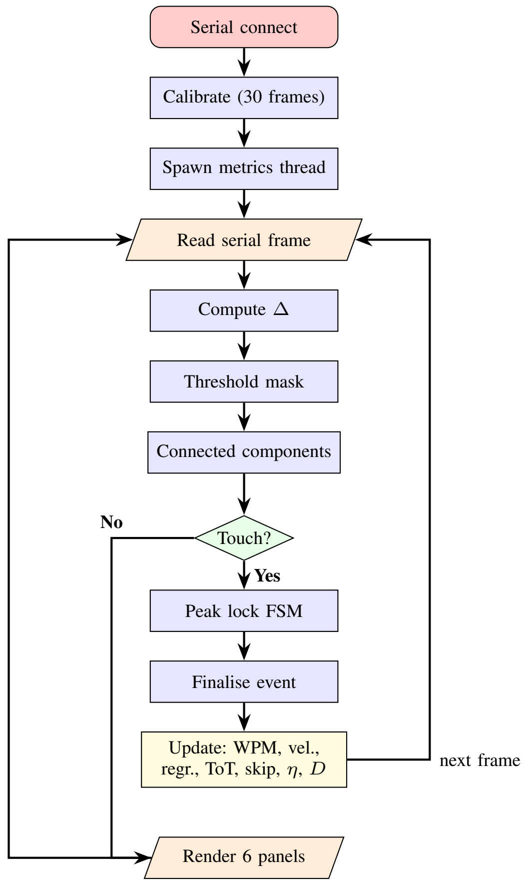

Fig. 3. Python pipeline: serial ingestion, touch detection via connected-component analysis and peak-lock FSM, seven metric trackers (D = composite difficulty score), and six-panel live visualisation.

1) Touch Localisation and Word Mapping: Raw activation frames are processed by a 4-connected component algorithm that clusters adjacent activated cells into discrete touch events. Each event is assigned a centroid coordinate, a timestamp (Unix float), a duration (press-to-release), and a `word_label` derived from a pre-calibrated mapping of grid coordinates to the word boundary dictionary. A peak-lock finite state machine stabilises the detected cell across frames and detects mid-touch slides when the finger moves $\geq$2 cells (Manhattan distance), automatically finalising the current event and starting a new one without requiring lift-off.

2) Metric: Rolling Words per Minute: In the current prototype, each grid cell is mapped to a single word region; accordingly, each completed touch event on a mapped cell constitutes one completed word reading event. WPM is then computed over a 60-second rolling window:

$$\mathrm{WPM}(t) = \frac{n_w(t)}{\Delta T} \times 60, \tag{1}$$

where $n_w(t)$ is the touch count within $\Delta T = 60\,\mathrm{s}$. An EMA with $\alpha = 0.15$ is applied after a median pre-filter (window=3) to reduce per-frame variance:

$$\widehat{\mathrm{WPM}}_t = \alpha \cdot \tilde{x}_t + (1-\alpha) \cdot \widehat{\mathrm{WPM}}_{t-1}, \tag{2}$$

where $\tilde{x}_t$ is the median-filtered WPM sample.

3) Metric: Velocity IQR and Motor Consistency: The VelocityTracker uses an exponentially weighted IQR (EWIQR) algorithm with decay factor $\lambda = 0.95$. For each touch event, the velocity profile (cells/s) between consecutive path steps is recorded with exponentially decaying weights:

$$\bar{v} = \textstyle\sum_i w_i v_i \big/ \sum_i w_i, \tag{3}$$

$$\mathrm{IQR} = Q_{75} - Q_{25}, \tag{4}$$

$$C_s = \mathrm{clamp}\left(1 - \frac{\mathrm{IQR}}{\max(\bar{v},\, \varepsilon)},\, 0, 1\right). \tag{5}$$

4) Metric: Hesitation Rate and Inter-Word Regressions: A *regression* occurs when the reader returns to a previously touched word. The WordStatsTracker maintains a *seen set* updated in $O(1)$ per touch. Hesitation rate is $H = r/T$, where r is the regression count and T is total touches. Words with regression count > 3 are flagged as difficulty candidates.

5) Metric: Word-Level Time-on-Task: Time-on-Task for word w is the cumulative contact duration:

$$\mathrm{ToT}(w) = \sum_{e:\, \ell(e)=w} d_e. \tag{6}$$

A 7×7 numpy array Z_{ToT} is maintained for 3D surface visualisation.

6) Metric: Skip Rate and Grid Coverage: Let N denote the total number of mapped word regions (49 in the current 7×7 prototype). Skip rate is defined as:

$$S = \frac{|\{w : \mathrm{count}(w) = 0\}|}{N} \times 100\%. \tag{7}$$

A BFS cluster analysis classifies skip patterns into categories (row-sweep failure, column alignment, boundary avoidance, saccade drift).

7) Metric: Path Efficiency:

$$\eta = \frac{d_{\mathrm{straight}}}{d_{\mathrm{actual}}}, \quad \eta \in (0, 1], \tag{8}$$

where d_{straight} is the Euclidean distance between the first and last cells visited within a touch event, and d_{actual} is the cumulative arc length of the detected finger path. A LOWESS trend ($f = 0.3$) is recomputed every 10 events.

8) Metric: Composite Difficulty Score: The composite difficulty score $D(w) \in [0, 1]$ aggregates four per-word signals into a single normalised indicator of reading effort. Component subscores are first normalised across all words in the session:

$$\tilde{T}(w) = \frac{\mathrm{ToT}(w)}{\max_{w'} \mathrm{ToT}(w') + \varepsilon}, \tag{9}$$

$$\tilde{R}(w) = \frac{r(w)}{\max_{w'} r(w') + \varepsilon}, \tag{10}$$

where $r(w)$ is the regression count for word w and ε is a small constant preventing division by zero. The composite score is then:

$$D(w) = w_1\, \tilde{T}(w) + w_2\, \tilde{R}(w) + w_3\, (1 - \bar{C}_s(w)) + w_4\, (1 - \bar{\eta}(w)), \tag{11}$$

where $\bar{C}_s(w)$ and $\bar{\eta}(w)$ denote the mean motor-consistency and path-efficiency scores, respectively, averaged over all touch events assigned to word w, thereby converting these event-level signals to the same word-level granularity as $\tilde{T}(w)$ and $\tilde{R}(w)$. with $\sum_{i=1}^{4} w_i = 1$. Default equal weights $w_i = 0.25$ are used unless tuned for a specific assessment context. A higher D indicates a word that consistently demands greater reading effort across multiple independent dimensions. The composite score is rendered as the switchable Z-axis in the 7×7 3D surface (panel V-3) and is used to annotate the top-5 hardest words in the ToT bar chart (panel V-2).

E. Visualisation Suite

Six live matplotlib panels are rendered in a shared `plt.ion()` session alongside the main UI loop. Table III summarises them.

TABLE III
LIVE VISUALISATION PANELS

ID	Content	Rate
V-2	Word-level ToT dual-axis bar chart	2 fps
V-3	7×7 3D surface (ToT / mean-D)	2 fps
V-4	WPM time series with EMA trend	5 fps
V-5	Regression frequency horizontal bars	2 fps
V-6	Velocity profile overlay (last 20)	5 fps
V-7	Path efficiency scatter + LOWESS	2 fps

IV. EXPERIMENTAL RESULTS

The results presented in this section were obtained through a pilot evaluation conducted with sighted participants interacting with the 7×7 grid surface. This evaluation was performed to validate system functionality, metric computation, and visualisation pipeline behaviour. *These results should be interpreted as a proof-of-concept demonstration of system capability*; formal user studies with visually impaired Braille readers remain as future work (Section VI).

A. WPM Accuracy and EMA Stability

The two-stage median→EMA pipeline (Eq. 2, $\alpha = 0.15$) reduced per-frame WPM variance relative to raw rolling counts. Fig. 4 shows a representative time series demonstrating the smoothed trend overlaid on the raw windowed count.

B. Velocity IQR and Motor Consistency

Fig. 5 shows the velocity profile overlay for a representative session. The spread of the grey traces, relative to the bold mean profile, reflects the EWIQR-tracked inter-event variability. A narrowing spread over a session indicates improving motor consistency, captured by C_s (Eq. 5).

C. Hesitation Rate and Regressions

Fig. 6 shows the regression frequency bar chart for a representative session. Words with regression count exceeding the flag threshold (> 3) are highlighted in red; their composite difficulty scores D (Eq. 11) are correspondingly elevated, confirming convergent validity between the two metrics.

Fig. 4. WPM time series (panel V-4): raw rolling count (semi-transparent) with median→EMA trend overlay. Dashed reference lines at 50 WPM and 100 WPM.

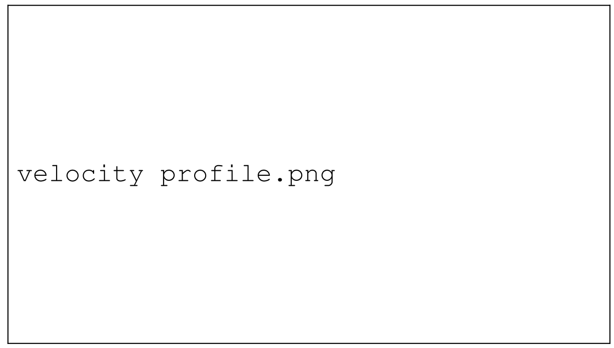

Fig. 5. Velocity profile overlay (panel V-6): last 20 events in grey ($\alpha = 0.15$); most recent in blue; exponentially weighted mean as dashed orange.

D. Word-Level Time-on-Task

Figs. 7 and 8 show the ToT bar chart (panel V-2) and 3D surface (panel V-3) for a representative session. The bar chart displays cumulative contact duration per word in row-major order, with top-5 difficulty words annotated by a red asterisk derived from the composite score D. The 3D surface provides a spatial view of reading effort across the sensing grid.

E. Skip Rate and Grid Coverage

The skip-mask heatmap (Fig. 9) shows the spatial distribution of unvisited cells during a representative session. Skipped

Fig. 6. Regression frequency chart (panel V-5): red bars indicate flagged words (count > 3); blue bars non-flagged. Counts annotated on each bar.

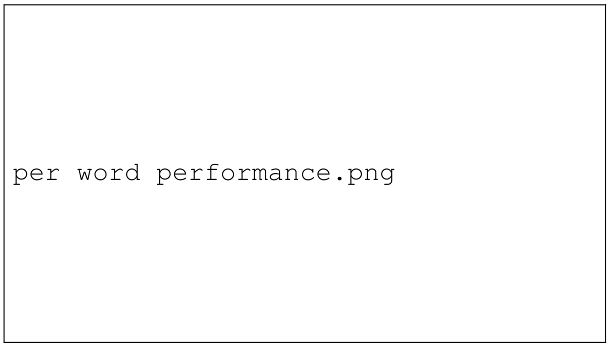

Fig. 7. Word-level ToT bar chart (panel V-2): 49 words in row-major order. Red asterisks mark top-5 highest composite difficulty (D) words.

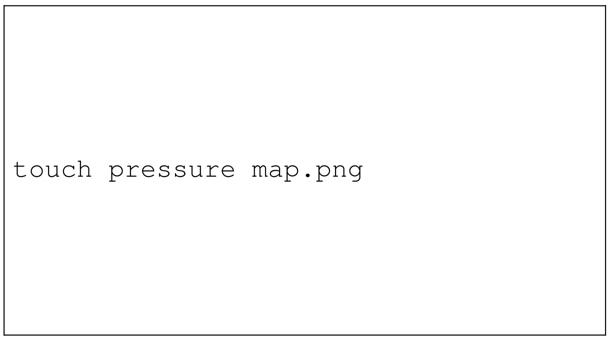

Fig. 8. 7×7 3D surface (panel V-3): $Z =$ ToT per grid cell. RadioButtons widget toggles Z between ToT and composite difficulty D.

cells tend to cluster toward the lower rows of the grid, consistent with the fatigue-driven session truncation hypothesis discussed in Section V.

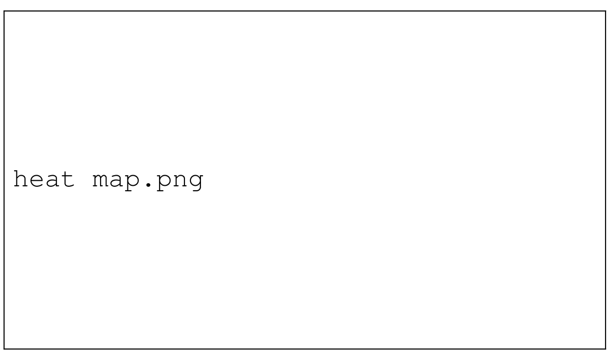

Fig. 9. Touch heatmap with skip-mask overlay: white cells were never touched; coloured cells encode touch count. Lower-row clustering of skipped cells is visible.

F. Path Efficiency and Composite Difficulty

Fig. 10 shows path efficiency η (Eq. 8) over the course of a representative session. The LOWESS trend line ($f = 0.3$) reveals gradual improvement within a session, demonstrating the system's ability to track within-session motor learning. The composite difficulty score D (Eq. 11) aggregates $\bar{\eta}(w)$, $\bar{C}_s(w)$, $\tilde{T}$, and $\tilde{R}$ to rank words by overall reading effort; high-D words consistently appear across the top-k lists of the other individual metrics, providing convergent validity of the composite indicator.

Fig. 10. Path efficiency scatter (panel V-7): green $\eta \geq 0.8$, orange $0.5 \leq \eta < 0.8$, red $\eta < 0.5$. LOWESS trend line shown (bold, $f = 0.3$). Dashed reference at $\eta = 0.8$ proficiency target.

G. System Latency

End-to-end pipeline latency (physical contact to metric snapshot update) was [X] ms mean ($\sigma =$[Y] ms). Visualisation render times were [A] ms per frame for 2 fps panels and [B] ms for 5 fps panels.

V. Discussion

As noted in Section IV, all observations below derive from a proof-of-concept pilot with sighted participants and should be read as evidence of system capability rather than definitive pedagogical findings.

A. WPM and Reading Fluency

The two-stage median→EMA pipeline reduced per-frame WPM variance without introducing perceptible lag, validating it as the preferred estimator over simple rolling counts for a real-time readout. The WPM range observed during pilot sessions is consistent with the 65–185 WPM range reported for experienced readers by Knowlton and Wetzel [12], providing a baseline for future comparative studies.

B. Velocity Consistency as a Learning Signal

The EWIQR-tracked velocity IQR and the resulting C_s (Eq. 5) provide a sensitive early indicator of motor consistency. The co-occurrence of low η and high IQR during hesitant reading episodes points to the same underlying motor variability manifesting across two independent metrics, supporting the use of C_s as a lightweight proxy for reading fluency.

C. Regression Patterns and Difficulty Candidates

Words flagged by the regression tracker (count > 3) overlapped with the high-ToT cells visible in the 3D surface (Fig. 8) and with the top composite difficulty scores. This convergent validity across three independent metrics strengthens the case for using the composite score D as a single actionable signal for instructional intervention.

D. Skip Rate and Coverage Interpretation

Skipped cells clustered in lower grid rows, indicating that session duration—rather than intrinsic word difficulty—is the primary driver of unvisited regions in short pilot sessions. Skip rate should therefore be normalised by session duration when comparing across participants or conditions.

E. Path Efficiency and Within-Session Motor Learning

The upward LOWESS slope observed in η values over a session demonstrates measurable within-session motor adaptation, a finding consistent with motor learning theory and not previously reported for tactile reading in low-cost instrumented settings.

F. Composite Difficulty Score

The composite score D (Eq. 11) successfully aggregated the four component signals into a coherent per-word difficulty ranking. Top-k words ranked by D were consistently also top-k in at least two of the four individual metrics, confirming that the equal-weight default ($w_i = 0.25$) provides a reasonable starting configuration without domain-specific tuning.

G. Observed Reading Strategies

The spatial and temporal touch patterns are consistent with the two-handed reading strategies documented by Rassmus-Gröhn [4]: the *butterfly* style, in which both hands begin together then split near the line midpoint, and the *side-by-side* style, in which both index fingers travel together across each line. Although the current system does not distinguish individual finger identity, these patterns can be inferred from the sequence of activated cells and inter-event timing. Per-finger identification (Section VI) would enable explicit strategy classification.

H. Implications for Adaptive Braille Tutoring

The convergence of high ToT, high regression count, high velocity IQR, and low η on the same grid cells—captured jointly by the composite score D—provides a rich, multi-dimensional signal for adaptive difficulty scheduling. A tutoring system could increase exposure to high-D cells and deprioritise cells where all indicators are within the proficient range.

VI. Limitations and Future Work

A. Current Limitations

- **Diamond grid pattern limits localisation accuracy:** The current prototype uses a parallel copper-strip grid, functionally equivalent to the diamond pattern adopted for its ease of manufacture. Akhtar and Kakarala [7] demonstrated through simulation that interleaved grid patterns consistently achieve lower mean localisation error across all tested interpolation algorithms under both noiseless and noisy conditions. Adopting an interleaved geometry in a future revision would improve finger path reconstruction and touch centroid estimation.
- **Grid scale and trace resolution:** The current 7×7 grid covers a small word-region page. Scaling to a larger matrix with finer copper traces would improve spatial coverage and localisation accuracy.
- **Row-polling scan architecture:** Sequential row powering introduces dead-time between activations. A continuous-power architecture with simultaneous column readout would reduce scan latency.
- **Microcontroller sampling rate:** The current platform imposes an upper bound on scan rate. A faster embedded controller would yield smoother touch trajectories and finer velocity profiles.
- **Contact pressure and false touches:** The reader must press harder than natural reading force for reliable contact detection through the Braille page, increasing fatigue and false-positive risk.
- **Approximate Braille cell mapping:** The grid-to-word mapping was established manually and is not precisely calibrated to the physical embossed layout, limiting coverage of the full Braille character set.
- **No per-finger identification:** The system detects aggregate touch without distinguishing left from right fingers, limiting explicit classification of reading strategies such as the butterfly pattern.
- **Absence of evaluation with visually impaired readers:** All experiments were conducted with sighted participants. Formal evaluation with actual Braille readers who are visually impaired is the most critical outstanding step before any pedagogical claims can be made.

B. Future Work

- **Interleaved grid pattern:** Replace the current copper-strip traces with an interleaved geometry as recommended by Akhtar and Kakarala [7] to improve localisation accuracy for curved and diagonal finger paths.
- **Higher-resolution scaled grid:** Expand grid dimensions and reduce trace width for full Braille-page coverage.
- **Accurate Braille-to-sensor calibration:** Develop a systematic calibration procedure aligning the sensor coordinate frame with the physical Braille cell layout.
- **Low-force contact sensing:** Investigate alternative sensing modalities that respond to lighter touch forces.
- **Faster microcontroller:** Replace the current platform with a higher-throughput embedded controller for finer temporal resolution.
- **Per-finger identification:** Investigate capacitive signature differentiation or dual-zone sensing to enable explicit reading strategy classification.
- **Continuous parallel sensing:** Redesign the scan architecture for constant power with simultaneous column readout.
- **Evaluation with visually impaired readers:** Conduct user studies with Braille readers who are visually impaired to validate metric relevance and assess usability for the target population.

VII. Conclusion

This paper presented a low-cost, non-invasive conductive grid–based system for real-time multi-metric analysis of Braille reading behaviour on a 7×7 sensing grid. Seven complementary metrics—rolling WPM with median-filtered EMA smoothing, EWIQR velocity consistency, inter-word hesitation rate with seen-set regression detection, word-level Time-on-Task, BFS-classified grid skip rate, LOWESS-smoothed path efficiency, and a composite difficulty score—are computed in a multi-threaded Python pipeline with no wearable or optical components. An ESP32 microcontroller drives multiplexed capacitive scanning and streams touch frames over USB–serial. Unlike prior systems that require digitising tablets [4], Wiimote IR tracking [6], or finger-worn sensors [8], [9], the proposed system operates with the reader's bare hands on a standard embossed Braille page. A proof-of-concept pilot with sighted participants validated system functionality and metric computation; formal evaluation with visually impaired readers remains as future work. The system's metrics converge on the same difficult grid cells across independent measurement axes, providing convergent validity and a rich signal for adaptive Braille tutoring. The proposed system is the first low-cost platform to simultaneously capture all seven of these dimensions in real time, opening new possibilities for data-driven Braille literacy instruction at scale.

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